

Question	Answer	Marks
7(a)	$R/R_T = 2.4/1.8$ or at 4.0°C , $R_T = 3.2\text{ k}\Omega$	C1
	hence $R/3.2 = 2.4/1.8$ $R = 4.3\text{ k}\Omega$	A1
7(b)	$R_T = 3.37\text{ k}\Omega$ or R_T is greater (than $3.2\text{ k}\Omega$)	B1
	$V^+ > V^-$	M1
	hence output is $+5.0\text{ V}$	A1

Question	Answer	Marks
7(c)	correct LED symbol	B1
	two diodes shown connected, in parallel and with opposite polarities, between V_{OUT} and earth	M1
	diodes labelled to show correct polarities consistent with (b) (G pointing from V_{OUT} to earth and B pointing from earth to V_{OUT} if (b) correct)	A1

Question	Answer	Marks
8(a)	force per unit current	B1